

Synopsis

Credit Card Fraud Detection using Machine Learning

Problem Id: 12

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Section- L

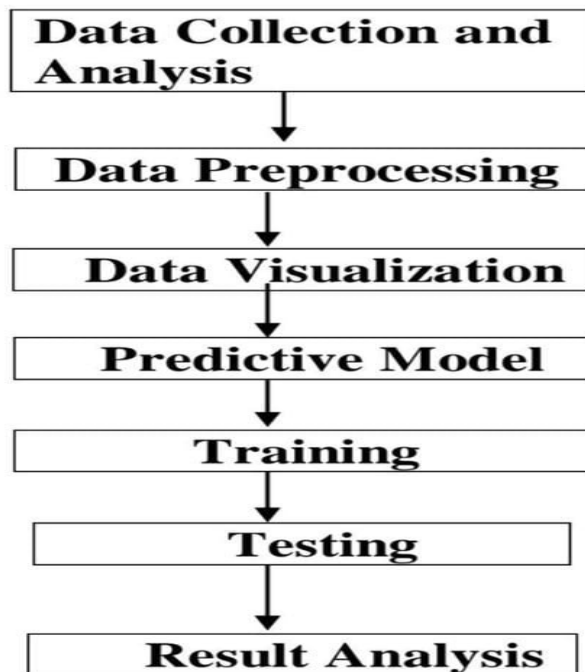
Problem Description:

In today's digital time, most of the people use credit cards or debit cards for different purposes like online shopping, paying bills, booking tickets, etc. This makes the work easy and fast, but at the same time it also increases the chances of fraud. Fraud means when someone uses the card details without permission and makes transactions. In our dataset, there are different columns such as transaction id, amount, time, merchant type, transaction frequency and class. The class column tells whether the transaction is **legitimate [normal](0)** or **fraudulent [fraud](1)**.

In this project, we are trying to solve this problem using machine learning techniques. We will build a system which can analyse the transaction data and predict whether a transaction is fraud or not. First, we will clean the dataset by checking missing values, removing duplicates and doing normalization wherever required.

Then, we will apply different machine learning algorithms such as Logistic Regression, KNN and SVM. After training the models, we will test them and evaluate their performance using different metrics to get better and more accurate results.

Steps of Implementation:



1. Data Collection

Collect a credit card fraud dataset which contains transaction details such as transaction id, amount, time, merchant type, transaction frequency and class.

2. Data Preprocessing

Handle missing values, remove duplicates, and normalize numerical features.

3. Feature Selection

Select important features that improve model accuracy.

4. Handling Imbalanced Data

Apply oversampling and undersampling techniques.

5. Model Selection

Use Logistic Regression, KNN, and SVM algorithms.

6. Training and Validation

Train models and validate using test data.

7. Evaluation

Evaluate using accuracy, precision, recall, confusion matrix and F1-score.

8. Final Model

Predict whether transaction is fraud (1) or normal (0).

Expected Modules/ Libraries to be used :

1. **Pandas:** The Pandas library in Python is primarily used for high-performance data manipulation and analysis, utilizing Series and DataFrames to manage structured data.
 2. **NumPy:** it is primarily used for array processing, linear algebra, statistical analysis, and as the foundational package for data science libraries like Pandas and Scikit-learn.
 3. **Matplotlib / Seaborn:** **Matplotlib** is open-source Python library designed for creating static, animated, and interactive visualizations like histograms, bar graphs, and line plots to understand data distribution and trends. **Seaborn** is a Python library built on Matplotlib designed for creating attractive, high-level statistical graphics and conducting exploratory data analysis (EDA).
 4. **Scikit-learn (sklearn):** **Scikit-learn** is used for implementing machine learning algorithms like Logistic Regression, KNN, and SVM. It also provides tools for data splitting and evaluation metrics.
 5. **Imbalanced-learn (imblearn):** A Python library used to handle imbalanced datasets using techniques like SMOTE and undersampling to improve model performance
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Expected Evaluation metrics and Plots to be used:

1. Accuracy

It is used in machine learning that measures the proportion of correct predictions (both true positives and true negatives) out of the total number of predictions made.

Accuracy formula = (Correct predictions) / (Total predictions).

2. Precision

It measures the accuracy of positive predictions, calculated as the ratio of true positives to all predicted positives

Precision formula = $TP / (TP + FP)$.

Where:

TP = True Positive. , TN = True Negative
FP = False Positive. , FN = False Negative

3. Recall (Sensitivity)

It measures the ability of the model to correctly identify all positive instances.

Recall formula= $TP / (TP + FN)$.

4. F1-Score

It is used for calculating the harmonic mean of precision and recall.

F1-score formula = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$.

5. Confusion Matrix

A confusion matrix is a table showing actual vs predicted values (TP, TN, FP, FN). It helps visualize model performance and types of errors.

	Predicted Positive	Predicted Negative
Actual Positive	TP (True Positive)	FN (False Negative)
Actual Negative	FP (False Positive)	TN (True Negative)

Plots Required :

1. Class Distribution Plot

- Shows the number of fraud vs non-fraud transactions.

- Helps identify data imbalance, which is common in fraud datasets.

2. Correlation Heatmap

- Displays relationships between features using colors.
- Helps in feature selection by finding strongly related variables.

3. Feature Distribution (Histogram/Boxplot)

- Shows how values of features are spread.
- Useful for detecting outliers and skewed data.

4. Confusion Matrix

- Table showing predicted vs actual results (TP, TN, FP, FN).
 - Helps understand types of errors made by the model.
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Required topics from the subject:

1. **Data Preprocessing:** It involves cleaning and transforming raw data into a suitable format for analysis. It includes handling missing values, removing duplicates, and normalizing data to improve model performance.
 2. **Classification Algorithms:** These algorithms are used to classify data into different categories. In this project, models like Logistic Regression, KNN, and SVM are used to detect fraudulent transactions.
 3. **Feature Selection:** Feature selection helps in identifying the most important features from the dataset. It reduces complexity and improves the accuracy and efficiency of the model.
 4. **Imbalanced Dataset Handling:** It deals with datasets where one class has significantly fewer samples than the other. Techniques like oversampling and undersampling are used to balance the data.
 5. **Evaluation Metrics:** These metrics are used to measure the performance of the model. Common metrics include accuracy, precision, recall, and F1-score.
 6. **Confusion Matrix:** A confusion matrix is a table used to evaluate classification models. It shows the number of correct and incorrect predictions made by the model.
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Platform Required:

Jupyter Notebook: An interactive development environment used to write and execute Python code. It allows easy visualization, testing, and step-by-step implementation of machine learning models.

Books and Link Sources:

1. Dataset created by students for the project
 2. <https://scikit-learn.org> (for understanding machine learning models)
 3. Hands-On Machine Learning with Scikit-Learn (reference book)
 4. <https://www.geeksforgeeks.org> (for basic concepts and examples)
 5. Class notes and study material provided in lectures.
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